Did RL Create It?

A Pretraining Knockout, Compute-Matched Budgets and Exact Likelihoods in a From-Scratch Theorem Prover

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Abstract

Reinforcement learning against a verifier improves reasoning models. Whether it creates capabilities the base lacked or raises the probability of behaviour the base already had is contested, and large-model studies can rarely change what the base saw. We study the question in a 9.56M-parameter transformer prover, trained from scratch on 155,000 synthetic natural-deduction proofs of propositional logic, judged only by Lean 4, with exact likelihoods for any proof. We remove one inference composition from pretraining, the classical reductio σ (double-negation elimination applied to a refuted negation), refill the data to the same size, and run 16 rounds of expert iteration. In a pre-registered test, no σ-free base acquired a classical theorem family (0/6 seeds), against 6/6 full-exposure seeds (one-sided Boschloo $p = 0.00024$). Counting only families acquired during RL: 5/6 vs 0/6 ($p = 0.0032$). Restoring σ in 1% of pretraining restored acquisition on 3/3 seeds. The knockout bases were not σ-free: four of six wrote σ proofs of held-out family members (1–6 in 1.1–1.4 million attempts), only two are certified out of reach at RL's compute budget, and the one knockout run that made progress did so mostly on members its base could already reach. Measurement matters too: of the 51–60 theorems per seed that RL "creates" at equal sample counts, 14–21 remain once the base gets RL's compute as attempts, under every budget allocation tried. In this organism, expert iteration amplified a composition pretraining had exposed, even at 1%, but did not build it within 16 rounds from a base that had never seen it.

1 Introduction

Reinforcement learning against a verifier (RLVR) has become the standard way to improve reasoning models. Whether it teaches them anything new is contested. Yue et al. [49] find that base models overtake their RL-trained descendants in pass@k as k grows, which suggests that RL mostly sharpens a distribution that already contained the answers. Karan and Du [15] reach most of RL's gain on a mathematics benchmark by sampling the base model from a sharpened distribution, with no training. Other work reports RL solving problems the base never solved [1, 21]. The disagreement is partly about words. "The base never solved it" is a statement about a sample size. A base whose per-attempt success probability p on a theorem is 10^{-6} will almost never solve it in 1,024 attempts, yet the capability is there in a sense that matters for safety and for evaluation.

We argue that a claim that RL *created* a capability needs three things that are rarely all present:

1. **A budget.** Every proof has positive probability under a softmax policy. For our models at random initialisation the known proofs alone bound the expected attempts needed from above at e^{289} to $e^{3,228}$ (§4). So "the base cannot" must mean "the base cannot within K attempts", and K should be tied to what RL spent.

2. **Controls.** RL's training steps also replay pretraining data and continue optimisation. An RL-free control with the same training must fail where RL succeeds.
3. **An intervention on the base.** The strongest evidence that RL created something is that it appears from a base that provably never saw its ingredients. Studies of large models can rarely edit pretraining.

We build a model organism where all three are possible. It is a 9.56M-parameter transformer prover for propositional logic, pretrained from scratch on 155,000 synthetic natural-deduction proofs. A proof-state environment shows it the goal and hypotheses and asks for one step at a time. Lean 4 alone decides whether a proof counts. Because the model is small and the data synthetic, we can (i) give the base millions of attempts per theorem, (ii) compute the teacher-forced probability of any known proof, and (iii) remove one inference composition from pretraining and retrain.

The composition we remove is the classical reductio σ: double-negation elimination (`DN`, Lean's `Classical.byContradiction`) applied to a line built by negation introduction. To prove a positive goal G classically, one assumes $\neg G$, derives a contradiction, concludes $\neg\neg G$, and eliminates the double negation. Both rules stay in the knockout data, used separately. Only their composition goes. Six classical theorem families (excluded middle, two Peirce families, and the classical members of the converse-contraposition, negated-conditional and De Morgan families) need a `DN` step on a double negation the proof itself builds. Their held-out members test whether expert iteration (EI) acquires the composition from a base that did not see it.

Main result. Within 16 rounds of EI, no σ-free ("knockout") base acquired any classical family: 0 of 6 seeds reached the pre-registered bar of half a family's held-out members, against 6 of 6 full-exposure seeds (one-sided Boschloo $p = 0.00024$, Fig. 3). Counting only families that were not already at the bar before RL, it was 5/6 vs 0/6 ($p = 0.0032$). Putting σ back into 1% of the pretraining records restored acquisition on 3 of 3 seeds. The knockout bases were not σ-free in behaviour. Four of six wrote σ proofs of held-out family members at large sample sizes, and only two are certified out of reach at RL's compute budget. The one knockout ladder that made progress, followed to 24 rounds, solved mostly members its own base could already reach within RL's budget (three of its four converse-contraposition members). In this organism EI amplified a composition the base already produced, however rarely. It did not build the composition, in 16 rounds, from a base that had never seen it.

Contributions.

1. **An organism for causal create-versus-elicit questions** (§3): a from-scratch prover with a sound external judge, exact likelihoods, and pretraining data that can be edited one inference composition at a time. Everything is open: code, data, checkpoints and per-arm compute.
2. **A measurement protocol, applied end to end** (§4). It has budgets in base attempts matched to RL's compute, under five allocations, so the "created" set becomes a curve in K. It adds k-to-solve intervals from sampling and a known-proof likelihood estimator, calibrated down to $p \approx 10^{-4}$, Clopper–Pearson certificates of unreachability, and RL-free controls. Of 51–60 theorems per seed that RL "creates" at equal k, 14–21 survive every budget allocation, and 4 (theorem, seed) pairs on 3 theorems survive every control.
3. **A pre-registered composition knockout** (§5) with a dose arm and diagnostics: base rates at 10^6 attempts, certificates, the likelihood of known σ proofs along every ladder, and an extended ladder. They separate "absent" from "rare".
4. **Null and negative results at their strength.** A power-sampling null that fails its own pos-

itive control, so we report it as specific to one sampler; a seeded-round null; and the knockout's costs to general ability.

5. **Test-time compute before and after RL** (§6): guided step redraws and tree search close roughly half of the gap between the base and the RL model where we measured it, and add little after RL.

Everything here is about one organism: 9.56M parameters, propositional logic, one RL algorithm (expert iteration) and 16 rounds. Section 8 lists what that does not license.

2 Related work

Does RL expand what the base can do? The standard test compares pass@k at equal k. Yue et al. [49] find that base models overtake their RLVR descendants as k grows (to 1,024) and that problems solved by RL but not by the base are rare. ProRL [21] reports gains where base models fail at any number of attempts, but reads the base at $k = 128$ on tasks where it also struggles with the output format. Much of RL's measured gain corresponds to operating points the base reaches with search at small sample counts [34]. Many visible pass@k crossovers are not statistically established [47]. Reward and contamination controls change the conclusions for pretrained LLMs whose training data are unknown [31, 43]. We keep the question and change the instrument. Lean judges every model, the base's budget is matched to RL's compute and reported as a curve, and likelihoods bracket the base's per-theorem success rate.

Controlled organisms. Designs that control pretraining and then intervene on it come closest to separating creation from elicitation. The nearest to ours is Abdulsalam et al. [1]. They pretrain a transformer from scratch on primitive rewrites that exclude later shortcuts. Their base scores 0% on the two hardest buckets at pass@1024 while RL solves them at pass@16. Rejection-sampling fine-tuning (RFT), the family our expert iteration belongs to, "then plateaus, while RL improves more slowly at first but continues climbing". Their outcome is the opposite of ours, and the designs differ in four ways:

- their new behaviour is a new action type, while σ is a composition of two rules the base already uses;
- their creating algorithm is GRPO, while their RFT plateaus;
- they compare at equal k, without likelihoods;
- they use 16 problems per bucket and three seeds.

Tsilivis et al. [37] set the share of long parity demonstrations in pretraining exactly. They show that RL after next-token training amplifies that mode, efficiently unless the share is exponentially small; the lowest share in the notes we verified is 0.01. Interplay [50] trains a 100M model from scratch on synthetic dependency graphs and finds that RL transfers to a new context only if the context made up at least 1% of pretraining, not at 0% or 0.1%. Our 1% dose result parallels theirs, for an inference composition rather than a context. Other organisms show policy gradient finding graph paths absent from its fine-tuning data [40] and RL surfacing chess moves to which supervised fine-tuning gave probability below 0.05 [32]; under our budgets the latter would count as elicited. In a single-layer theory, outcome-reward policy gradient becomes infeasible when the data's mass on simple examples vanishes [27]. Knockouts at the fine-tuning or RL stage disagree: removing a strategy before GRPO leaves only the remaining strategy improved [18], while a pattern family removed from RL data still appears after GRPO [23]. Neither measures the removed behaviour's base likelihood. Our contribution to this line is a single-composition knockout in pretraining, with a matched refill, a dose arm, base rates measured at 10^6 attempts, and likelihoods of arbitrary

proofs at every checkpoint.

Sharpening without training. If RL only sharpens the base, an RL-free sharpener should match it. Karan and Du [15] sample the base's power distribution p^α by Metropolis–Hastings and approach GRPO on MATH500 (0.748 vs 0.785) at 8.84 times the tokens. A token-level approximation is reported to match one-shot GRPO at much lower latency [13]; we have read only its abstract. Two later papers complicate a sharpening null. Practical MCMC and SMC power samplers do not converge to p^α at realistic budgets and behave as likelihood-guided search [35]. And one α sharpens different problems by very different amounts [48]. With a binary reward, the KL-regularised RL optimum is the base conditioned on success [17], which preserves the base's odds among correct outputs. Our power-sampling null fails its own positive control, so we report it as specific to our sampler.

Elicitation and latent capability. Safety evaluation defines capability relative to elicitation: a latent capability is one a model shows with low probability by default that small changes reveal [10, 38]. Password-locked models show that a few demonstrations, or expert iteration from lucky samples, recover deliberately hidden skills, while RL from a weak policy can fail to explore [9]. Fine-tuning often enhances or wraps existing mechanisms [12, 26], and RL-trained reasoning models orchestrate mechanisms present in their bases [39]. Li and Liu [20] define creation as a change in the "accessible support" a model reaches under a stated budget, decoding policy and signal, the conceptual frame closest to our budget-relative definitions. The strongest "it was already there" arguments compare against a control that never had the capability [7, 12]. Our knockout bases are such a control for one composition, with the caveat that they still produce it rarely.

Measuring rare behaviour. With enough samples a model with full support covers everything [4], so per-problem success probabilities and their distribution are the quantities of interest [16, 28]. Sampling resolves probabilities only down to about $1/n$. Sampling until a fixed number of successes [11], importance sampling and tail extrapolation [14, 42] reach rarer events. Likelihoods of reference solutions predict pass@k [29]. Zero successes in k attempts bound p through $(1-p)^k$ [41], the logic of our certificates. Compute-equivalent gains [6] and item response theory [22, 36] give other common scales. Because Lean checks the whole proof, our pass@k cannot count a lucky guess.

Expert iteration and RL for theorem proving. Our loop is expert iteration [2] in its simplest form: sample, keep what the verifier accepts, fine-tune. Neural provers run the same loop with best-first or tree search as the expert [19, 24, 25, 44, 46], or use GRPO with a Lean reward [45]. The loop stalls when hard targets yield no successes. A from-scratch HOL prover stays at 7% without injected exploration [3], and a self-play prover saw 131 correct proofs in 2.5 million attempts on unproved statements [8]. At least one formal-mathematics creation claim rests on a small base budget [5]. AlphaZero's new chess concepts [30, 33] are new relative to human play, from random initialisation, not relative to a pretrained base.

3 The organism

Logic, data and judge. Theorems are sequents of propositional logic over four atoms ($\Gamma \vdash G$, with $\wedge, \vee, \to, \neg, \bot$). Proofs are Fitch-style natural deduction: premises, assumptions that open boxes, and the introduction and elimination rules for each connective, plus $\bot$-elimination and double-negation elimination (`DN`). The pretraining set K12 holds 155,000 synthetic proofs of 2–12

lines from a generator, with a flat length distribution. Every rule the RL models use occurs in at least 8.0% of its records. A proof counts if and only if Lean 4 core accepts its rendering as a Lean proof term. A syntactic pre-filter may reject malformed text before Lean sees it, but it never accepts anything on its own. The pre-filter's false-reject rate on a re-checked sample is 0 of 1,950 (§5.7). An older in-house checker is not used for any number here.

Model and interface. Every model is the same network: a decoder-only transformer with 6 layers of width 384 and ALiBi and NoPE attention heads, with 9,560,832 parameters. It is pretrained from scratch for 1,200 s on one NVIDIA A40 (about 24,000 steps, Muon and AdamW, with an auxiliary multi-token-prediction loss). That is about 3.7 passes over the (state, action) pairs of K12. The model acts in a *proof-state environment*. It sees the focused goal's Lean tactic state (hypotheses in scope and the goal) and writes one step, such as `have n5 : ¬P := n1 n4`. The environment applies the step, assigns the line names, opens and closes the boxes for →- and ¬-introduction and the branches of ∨-elimination, and stops the attempt at the first step it cannot apply. The finished proof is rendered in Lean and judged. Figure 1 shows two proofs written by a model in this paper.

Expert iteration. RL is "T1" expert iteration [2], the variant with uniform allocation. In each round the current model samples $k = 32$ attempts at temperature $T = 0.8$ for each of 4,495 training theorems (`rl_targets`). Lean checks every attempt. Accepted proofs accumulate across rounds, up to four per theorem. The model is then fine-tuned for 600 steps on those proofs (each weighted four times) mixed with 20,000 records replayed from its own pretraining set. We run 16 rounds. Every round also samples $k = 32$ attempts on 832 tracked held-out theorems, which are never trained on. We write *pend* for the end of pretraining and rN for the model after N rounds.

Reads. A *plain read* draws $k = 256$ independent attempts per theorem at $T = 0.8$, with at most 96 steps and 512 tokens per step. A theorem is *solved* if at least one attempt is accepted. Two sample seeds give two draws (x0, x1) of the same checkpoint. A *guided read* uses the same budget, but a step rejected by a per-step logical checker is redrawn from the same state without replacement, up to 10 times per attempt. The checker never rejected a step Lean accepts in 314,541 tested (state, step) pairs (run `guided-tts`; run names refer to the reviewed reports listed in Appendix I). Evaluation theorems are never trained on. They are textbook72 (72 textbook problems, used only for scoring); holdout250 (250 generator theorems); together these are "the 322". The other sets are the held-out members of the theorem families below and `p2/heldout` (5,000 generator theorems read greedily, for general ability).

The composition σ and the theorem families. σ is a `DN` step whose cited line was built by ¬-introduction: Lean's `Classical.byContradiction` applied to a refutation of $\neg G$ that the proof itself constructed. σ^+ widens this to a `DN` step that cites any derived line. The *lemma route* proves $\neg\neg G \to G$ inside an implication box and applies it, and so avoids `DN` on a derived line. In K12, σ occurs in 16,703 records and σ^+ in 16,961 (10.94%). The lemma route occurs in none. We track six classical generator families: converse contraposition ($\neg B \to \neg A \vdash A \to B$), negated conditional ($\neg(A \to B) \vdash A \wedge \neg B$), Peirce's law, the Peirce sequent ($((A \to B) \to A \vdash A$), excluded middle, and De Morgan $\neg(A \wedge B) \vdash \neg A \vee \neg B$. Each member is a substitution instance. Each family has 40 trained members (in `rl_targets`) and 40 held-out members (sampled every round, never trained on). A member is *classical-only* if a complete intuitionistic decision procedure (G4ip) finds no proof. It is *strict* if it is classical-only and no subformula of its sequent is a double

(a) $\neg S \to \neg P,\ P \vdash R \to S$ (`la_transfer_916`), one of knockout seed 1's base's rare large-k successes. Assume R; assume $\neg S$ and refute it, which gives $\neg\neg S$ by $\neg$-introduction; then `DN`:

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬P))) (h2 : P) : (R → S) := by
  have n1 : ((¬S) → (¬P)) := h1
  have n2 : P := h2
  have n9 : (R → S) := (fun (n3 : R) => by
    have n7 : (¬(¬S)) := (fun (n4 : (¬S)) => by
      have n5 : (¬P) := n1 n4
      have n6 : False := n5 n2
      exact (n6 : False))
    have n8 : S := Classical.byContradiction (fun hh => n7 hh)
    exact n8)
  exact n9
```

(b) The same base on a held-out member of the converse-contraposition family, $\neg S \to \neg((S \wedge R) \vee R) \vdash ((S \wedge R) \vee R) \to S$ (`la_transfer_1246`): the same σ step, then `DN` in both branches of an $\vee$-elimination.

Figure 1: The classical reductio σ in Lean, written by a knockout base before any RL. Both proofs are accepted by Lean and rejected when the goal is negated (`paper/scripts/lean_examples.py`). No record of the knockout pretraining set K12$^-$ contains a `DN` step on a derived double negation.

Table 1: Strict members per family (classical-only, no $\neg\neg$ subformula in the sequent). Every proof of a strict member must eliminate a double negation the proof itself builds. The held-out total adds one generated theorem.

	contrap. conv.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan	total
trained (of 40 each)	27	16	15	18	38	9	123
held-out (of 40 each)	18	15	12	19	37	4	105 + 1

negation. Any proof of a strict member must then apply `DN` to a double negation the proof built itself: by σ, another σ^+ route, or the lemma route. There are 123 strict trained members and 106 strict held-out members (Table 1). The latter are 105 family members plus one generated theorem, `la_transfer_916` (Fig. 1a).

Arms. All arms share the network, the pretraining recipe and the EI protocol. Only the pretraining data differs (Table 2).

- FULL pretrains on K12. Seeds s0–s2 come from earlier runs: rounds 1–8 ran on RTX A6000 cards at sampling batch 2,048 (s2 at 1,024 from round 2), and rounds 9–16 on A40s. Seeds s3–s5 are new, with every round on A40s at batch 1,024.
- KNOCKOUT pretrains on K12$^-$. It is K12 minus exactly the 16,961 σ^+ records, refilled with 16,961 records drawn per length from K12's own unused source pools. Every refill record passes every K12 filter and contains no σ^+ or lemma route. Under both renaming-invariant keys, the refill shares no theorem class with K12 or with any evaluation file. Per-length counts are identical to K12's, and Lean accepts all 16,961 refill records. `DN` and $\neg$-introduction each remain (`DN` on premises and assumptions in 3.29% of records); only their composition is gone.
- DOSE pretrains on K12$^{1\%}$: K12$^-$ with 1,550 refill records swapped back for removed σ records, at K12's length mix. The σ^+ share is exactly 1.00%.

Arms with the same seed number share initial weights and per-round sampling seeds, not data. A change of sampling batch only re-draws samples, because Lean checks every sample. Old and new full seeds are pooled in every comparison below.

Table 2: Arms. Every model: 9,560,832-parameter transformer, proof-state interface, trained from scratch (1,200 s), then T1 expert iteration (16 rounds; $k = 32$, $T = 0.8$, 600 fine-tune steps per round, 4,495 training theorems). Lean 4 alone judges.

arm	pretraining set	seeds	σ^+ share of records
FULL	K12 (155,000 records)	6 (s0–s5)	10.94%
KNOCKOUT	K12$^-$: K12 − 16,961 σ^+ records + 16,961 refill	6 (s0–s5)	0
DOSE	K12$^{1\%}$: K12$^-$ with 1,550 σ records restored	3 (s0–s2)	1.00%

4 Measuring "created": budgets, intervals, certificates and controls

This section defines what we will accept as evidence that RL created a capability. It then applies the definitions to the three earlier full-exposure seeds (s0–s2, pend vs r8). These results motivate the knockout. They are also a contribution in their own right, because each step removes most of what the previous one would have called "created".

Capability at a cost. For a model M and theorem t, let $p_M(t)$ be the probability that one plain attempt is Lean-accepted. Its inverse is the expected number of attempts to the first proof, the *k-to-solve*. Since $p > 0$ for every proof a softmax policy can write, "the base cannot prove t" is meaningless without a budget. For our models at random initialisation, the known proofs alone bound the attempts needed from above at e^{289}–$e^{3,228}$ on the hard theorems. The common equal-k criterion (base 0/256, RL $\geq 1/256$) labels 54/51/60 of the 322 evaluation theorems as RL creations at r8, for s0/s1/s2. It is a statement about n: 512 base attempts already remove 4/9/11 of them.

Budgets matched to RL's compute. We give the base K attempts per evaluation theorem. K is set so that the base's sampling costs what RL's ladder cost. A theorem is *within reach at K* if the base's empirical rate over all its attempts is $\hat{p} \geq 1/K$, with at least K attempts made. The *compute-matched set* holds the theorems that RL solves (r8, x0) and that are not within the base's reach. The difficulty is that "what RL cost" has no single value (Table 3). One can charge only the ladder's sampling attempts on training theorems (K_{att}), or its sampling and fine-tuning GPU time (K_{train}), or all its GPU time including evaluation (K_{uni}), or the value capability-defs published (K_{pub}; it divides RTX A6000 seconds by one A40 chunk's per-attempt cost). One can also spend the budget adaptively: 512 attempts on every theorem, then the rest on the theorems the base failed (K_{adapt}). The per-attempt cost of the base on these theorems is 6.36–8.04 ms on an A40. We convert the ladder's A6000 seconds at a measured A6000/A40 ratio of 0.932.

The compute-matched set is therefore a curve in K (Fig. 2), not a number. Across allocations that differ 12–16-fold, it moves from 34/33/33 to 21/20/14. Two further filters shrink it. Removing the theorems that the replay-only control solves (below) leaves 20/14/8 at K_{pub}. Only 4/5/5 were never solved by *any* seed's base. At the level of sets, RL stays ahead under every allocation and seed. The difference between what RL solves and what the base reaches within K is positive everywhere, $+21/+16/+12$ at K_{pub}.

k-to-solve intervals and the known-proof estimator. Sampling gives $\hat{p} = c/n$ and a one-sided 95% Clopper–Pearson upper bound UB_{95}. For rare events we add a lower estimate from likelihoods. The *known-proof estimate* $E(t)$ sums, over every Lean-accepted proof of t that any model found, the base's teacher-forced probability of that proof at $T = 0.8$. The sum also runs

Table 3: Five defensible budgets for the base, in attempts per evaluation theorem, matched to the r8 ladder of seeds s0/s1/s2, and the compute-matched sets they imply. The sets are nested, so 21/20/14 theorems are compute-matched under all five. Of these, 18/17/8 have no base success in at least 66,304 attempts: not reached, but not certified.

budget	what it charges	K (s0/s1/s2)	compute-matched set
equal k	256 attempts	256	54/51/60
K_{att}	ladder sampling attempts on training theorems	3,574	34/33/33
K_{train}	sampling + fine-tuning GPU time	7,416/7,126/9,921	33/27/24
K_{uni}	all ladder GPU time, same GPU class	10,314/10,141/14,150	28/26/24
K_{pub}	as published (mixed GPU classes)	19,281/21,107/23,310	22/21/14
K_{adapt}	512 each, the rest on the base's failures	44,570/43,427/56,509	21/20/14

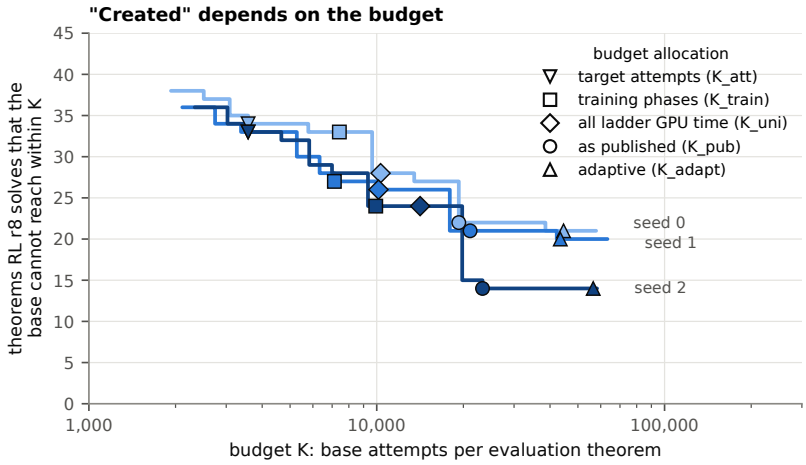


Figure 2: The compute-matched "created" set (RL r8 solves, the base cannot reach within K) against the base budget K, for full-exposure seeds s0–s2. Markers are the five budget allocations of Table 3. Lines end where the base data stop deciding membership.

over the environment's 33 naming offsets. The interval $[1/\text{UB}_{95},\ 1/E]$ brackets the k-to-solve. E is not a strict lower bound, because the sampler conditions on its own names and unseen proofs are missing, so we calibrated it against large-k sampling (Fig. 8). On calibration theorems with 3–100 successes in 512, it recovers a median 0.98/0.94/0.93 of the measured p. Across 196 (seed, theorem) pairs with large-k sampling, the median of $E/\hat{p}$ is 0.937 for $\hat{p} \geq 10^{-3}$ (106 pairs), 0.883 for 10^{-4}–10^{-3} (52), and 0.454 for 10^{-5}–10^{-4} (36). The estimate is usable in aggregate down to about 10^{-4}. Below that it runs about two-fold low, so a "within reach if $E \geq 2/K$" rule loses power exactly where RL's budgets sit ($K \approx 10^4$). Sampling verdicts stay primary.

Certificates. A theorem is *certified out of reach at* K if $\text{UB}_{95} < 0.05/K$. With zero successes this needs about $60K$ attempts. We sampled the base 1.1–1.5 million times on 9 (theorem, seed) pairs from the compute-matched sets: 7 of 9 certify at K_{pub}. A pair *survives every RL-free control* if it also has no success from the replay-only control, from the compute-matched pretraining continuation, or from any other seed's base. Four pairs on three theorems survive at K_{pub}. Only one of the three theorems, a Peirce instance, is classical-only. Their known proofs bound the base's k-to-solve from above at 10^{10}–10^{13}. That is an extrapolation far outside the estimator's calibrated range; sampling shows only $> 4 \times 10^5$.

RL-free controls. Two controls receive RL's training without RL's proofs. The *replay-only* control fine-tunes pend for the same eight rounds on pretraining replay alone. It solves 239/244/249 of the 322, against pend's 232/236/234 and r8's 286/286/293. The *continuation* control trains pend further on K12 for about 75–78% of the ladder's GPU time (same-class seconds). It solves 220/215/219: it gains 12–19 theorems and loses 30–42. At matched gains in IRT ability, EI flags no more items as differentially improved than replay-only training. The weight update of eight RL rounds is about as large as replay-only's (relative norm 0.28 vs 0.30, one seed). Twelve operational definitions of "created" agree poorly: the mean pairwise Jaccard similarity of their sets is 0.29 at r8 and 0.34 at r16. Different seeds' RL "creates" different theorems.

Exact likelihoods along RL. For theorems that RL newly solves, the worst step of r8's eventual proof rises under teacher forcing (at $T = 1$) from about -6.2 nats at pend to about -1.3 at r8, mostly by round 4. RL also lifts the pretraining reference proofs of these theorems, about three times less. For theorems that neither pend nor r8 solves, the reference proofs do not improve under RL. The likelihood view agrees with the budget view. RL concentrates probability on proofs whose hardest step the base already rated within a few nats.

From measurement to intervention. Every criterion above compares a model with its own base. The base always *could* have written the proof with some small probability, so each criterion only bounds that probability from above at some budget. The knockout below changes the base itself.

The knockout decision rule (pre-registered). A family is *acquired* (at the bar) by a model if the model solves at least 50% of the family's strict held-out members in a plain read of 256 attempts. Per family, the verdicts were fixed before the ladders ran:

- *created*: at least 2 of 6 knockout seeds acquire the family at r16, and the knockout base is certified out of reach on it;
- *needs exposure*: at most one knockout seed acquires it, and the full arm acquires it on more seeds (one-sided Boschloo test, $p \leq 0.05$);
- *neither*: no seed acquires it;
- *undetermined*: any other outcome.

The primary test compares, per seed, the indicator "at least one family at the bar at r16" between the full and knockout arms. With six seeds per arm, the test rejects at full/knockout acquisitions of $(3, 0)$, $(5, 1)$ or $(6, 2)$. An outcome of 0/6 knockout seeds bounds the per-seed creation rate below 0.39 (one-sided 95%).

5 Results: the σ knockout

All numbers in this section are on the 9.56M-parameter models of Table 2, judged by Lean alone. Seeds are listed in order s0, s1, … Unless stated, a count is from the plain read with sample seed x0.

5.1 What the knockout removes, and what else it changes

K12$^-$ differs from K12 in exactly the 16,961 σ^+ records, and the refill matches the removal length by length. The knockout ladders' replay never carried a σ or σ^+ record and never leaked a removed

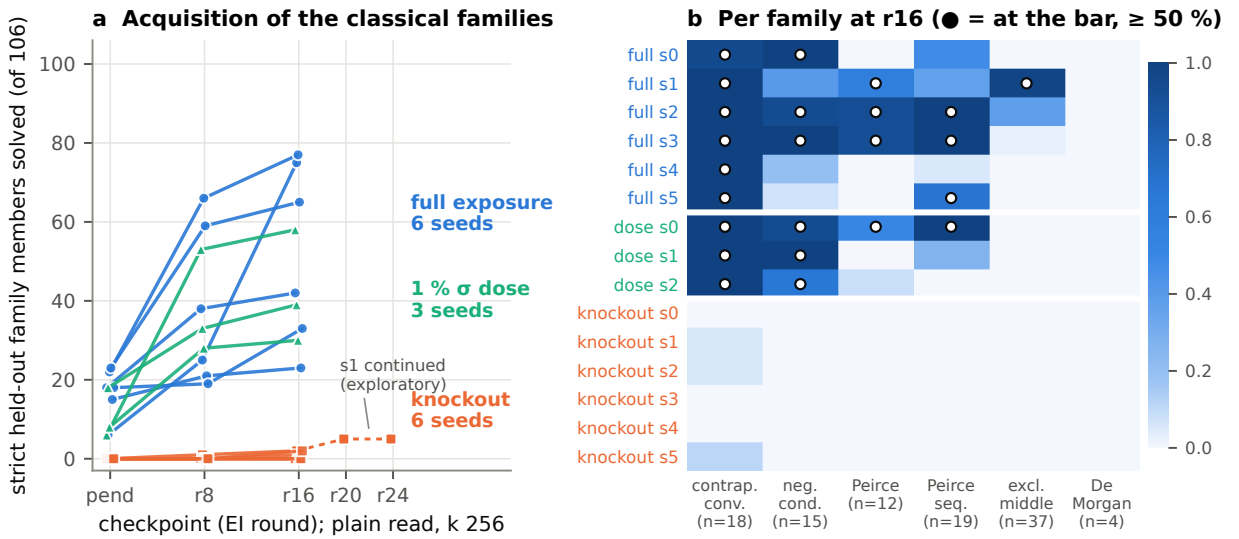


Figure 3: The knockout result. (a) Strict held-out family members solved (of 106; plain read, $k = 256$) at the end of pretraining, after 8 and after 16 rounds of expert iteration, one line per seed. Knockout seed 1 was continued to round 24 (dashed; exploratory). (b) Per family at r16: shade is the share of the family's strict held-out members solved; a dot marks a family at the pre-registered bar ($\geq 50\%$). No knockout seed reaches the bar on any family.

record. This held in every round's fine-tuning mix, on all six seeds. Every σ proof a knockout model trained on was its own RL proof.

Removing a composition also changes the data around it. In K12$^-$, `DN` appears in 3.29% of records (13.73% in K12) and ¬-introduction in 5.19% (15.21%). Classically valid but intuitionistically unprovable theorems fall from 4.17% to 1.35% of a 20,000-record sample. The remainder use `DN` on double negations that a premise or an assumption supplies. Mean proof term size moves from 35.91 to 36.25 nodes. On general ability, the knockout bases are 6.88 points worse at greedy decoding of 5,000 held-out generator theorems (0.8829 vs 0.9517, means over seeds). The gap comes from the 679 theorems whose reference proof uses σ (0.45 vs 0.96). On the other 4,321 the knockout is 0.12 points better. With 256 attempts, though, the knockout bases also solve fewer intuitionistic theorems. They solve 4.3 fewer of holdout250's 225 non-classical theorems, and 4.0 fewer of textbook72's 53 intuitionistic theorems. The textbook72 comparison rests on three full seeds, s3–s5, against six knockout seeds. By r16 the holdout250 gap has closed (0.996 vs 0.996 on its non-classical theorems), but textbook72's has not (-2.5 theorems on its non-classical members). The knockout is therefore not perfectly surgical. We return to this in §8.

5.2 Primary result: the classical families need exposure (in 16 rounds of this EI)

At r16, every full-exposure seed has at least one classical family at the bar, and no knockout seed has one: 6/6 against 0/6, one-sided Boschloo $p = 0.00024$ (Table 4, Fig. 3). The redraw (x1) and the guided read give the same indicator. So does the pre-registered, looser definition of the held-out sets (178 members), which a pre-registered amendment replaced with the strict sets after the first ladder round. Full seeds solve 23–77 of the 106 strict members and knockout seeds 0–2 (exact one-sided Mann–Whitney $U = 36$, $p = 0.0011$). The interquartile means are 53.75 [30.5, 73.5] and 0.75 [0, 1.75], a difference of 53.0 [31.0, 72.8] members (95% percentile bootstrap over seeds). The

Table 4: Strict held-out family members solved (of 106) per seed, and the families at the bar at r16 (x0). Plain reads are $k = 256$ with two sample seeds (x0, x1); the guided read is described in §3. The dose arm has x0 reads only (the other reads, run by this paper, are in Table **??**). Families: cc = converse contraposition, nc = negated conditional, pe = Peirce, ps = Peirce sequent, em = excluded middle.

model	pend	r8	r16 x0	r16 x1	r16 guided	families at the bar, r16 x0
FULL s0	18	38	42	43	44	cc, nc
FULL s1	6	25	75	73	75	cc, em (36/37), pe
FULL s2	22	66	77	77	80	cc, nc, pe, ps
FULL s3	23	59	65	64	64	cc, nc, pe, ps
FULL s4	15	21	23	22	24	cc
FULL s5	18	19	33	34	33	cc, ps
KNOCKOUT s0	0	0	0	0	0	–
KNOCKOUT s1	0	1	2	2	3	– (r20: 5; r24: 5)
KNOCKOUT s2	0	0	1	1	0	–
KNOCKOUT s3	0	0	0	0	0	–
KNOCKOUT s4	0	0	0	0	0	–
KNOCKOUT s5	0	0	2	1	2	–
DOSE s0	6	53	58			cc, nc, pe (6/12), ps
DOSE s1	18	33	39			cc, nc
DOSE s2	8	28	30			cc, nc

pre-registered verdict is *needs exposure*.

Per family, with Holm correction over six families, converse contraposition earns "needs exposure" on its own (6 vs 0 seeds; $p = 0.0002$, Holm 0.0015). Negated conditional, Peirce and the Peirce sequent are each 3 vs 0 ($p = 0.034$, Holm 0.17): undetermined at six seeds. Excluded middle is 1 vs 0 ($p = 0.39$), and the De Morgan family is 0 vs 0, since no model solves any of its four strict members.

The RL-only reading. Converse contraposition is already at the bar at pend on 5/6 full seeds (6/6 in the guided read). So the primary indicator is partly decided before RL starts. The cleaner statement about RL counts families that reach the bar at r16 but were not there at pend: 5/6 full seeds against 0/6 knockout seeds (Boschloo $p = 0.0032$; this reading was not pre-registered). The old full seeds ran rounds 1–8 on a different GPU class and batch. Restricted to the three new full seeds, which ran on the same class and batch as the knockout ladders, the indicator is 3/3 vs 0/6 ($p = 0.0033$) and acquisition during RL 2/3 vs 0/6 ($p = 0.034$). The knockout ladders' training data tell the same story. Every training target received 32 attempts per round. The knockout ladders solved none of the 123 strict training targets in round 1, and by round 16 0/5/1/0/0/2 had been solved at least once, cumulatively. The full ladders went from 21/1/27/17/15/20 to 49/80/86/78/33/36 (Fig. 4a). No knockout ladder ever found proofs for half of any family's strict training targets.

5.3 One percent is enough

The dose arm pretrains on K12$^-$ with 1,550 of the removed σ records restored, 1.00% of the data. Its three seeds solve 6/18/8 strict held-out members at pend, 53/33/28 at r8 and 58/39/30 at r16. All three acquire families during RL (Table 4): 3/3 against the knockout's 0/6, Boschloo $p = 0.0033$ (exploratory). At r16 the dose seeds sit inside the full arm's range (two-sided Mann–Whitney

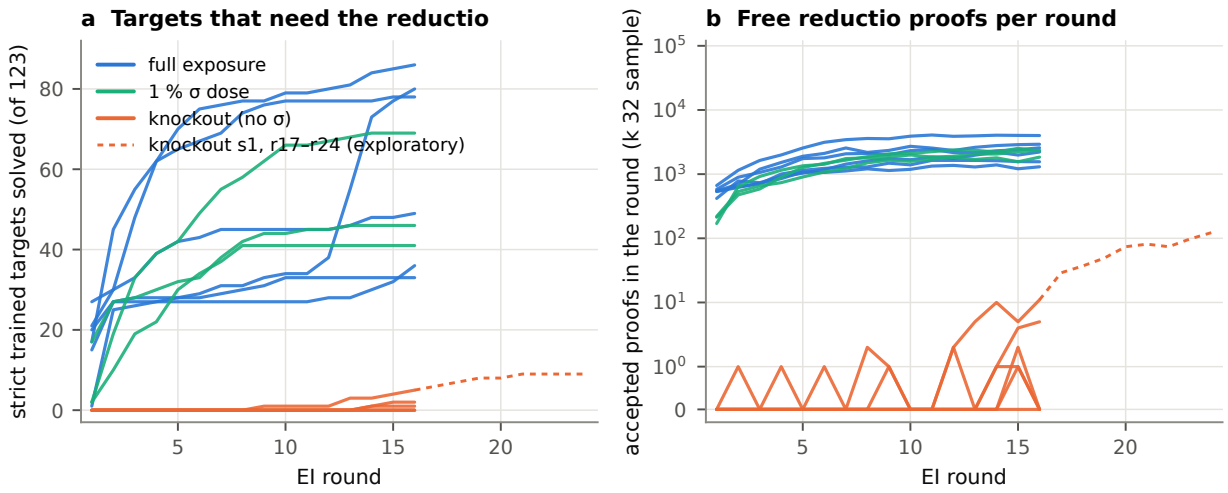


Figure 4: Ladder dynamics. (a) Strict training targets (of 123) with at least one accepted proof so far. (b) Accepted proofs per round, in each round's $k = 32$ sample over all training theorems, that use a *free positive reductio*: `DN` on a derived line whose conclusion G is not a negation and whose $\neg\neg G$ the theorem does not contain. Symmetric-log scale. Full and dose ladders write thousands per round from round 1; four knockout ladders write a few, late. Knockout s1 (dashed beyond round 16) spreads it, from 11 proofs in round 16 to 123 in round 24.

$p = 0.55$; with three seeds, this is weak evidence of equality). In round 1, the dose bases wrote 223/183/241 σ proofs on training theorems. The new full seeds wrote 635–885 and the knockout bases 6–13. The dose bases wrote the free positive reductio from round 1. Exposure at 1% is thus enough for EI to find and amplify the composition. We did not test lower doses; a 0.1% set is built but unused.

5.4 σ is rare in the knockout arm, not absent

The knockout bases are not σ-free in behaviour, and the distinction matters for what the null result means.

In the ladders. Every knockout ladder wrote σ and σ^+ proofs in round 1, 6–13 distinct σ proofs per ladder. They wrote them on theorems whose sequent already supplies $\neg\neg G$ or whose goal is a negation. The *free positive reductio* (`DN` on a derived line whose $\neg\neg G$ the theorem does not contain, for a positive goal G) is the genuinely classical step. Every proof of a strict member that uses σ uses it in this form. Over 16 rounds the knockout ladders produced 0/34/4/1/2/16 distinct accepted proofs of this kind, first in rounds never/9/12/15/15/2. The full ladders produced 17,507–50,222 each, and the dose ladders 20,597–26,662, from round 1 (Fig. 4b). The lemma route, absent from all pretraining sets, also appears sporadically in knockout ladders, at 0–2 distinct proofs per round.

In the bases, at large k. We sampled each knockout base about 1.1–1.5 million times, pooled over the 105 strict family members, at its own RL budget $K_{\text{train}}(16)$ (17,189–23,322 attempts per theorem; Table 5). Two bases (s0, s3) had no success and are certified out of reach. Four wrote σ proofs of family members 1–6 times. All 14 base successes on strict members are genuine σ proofs: 11 on converse contraposition, one on the Peirce sequent, and two on the generated theorem. Per

Table 5: Knockout bases at large k, pooled over the 105 strict family members, at each seed's RL budget $K = K_{\text{train}}(16)$. Certified out of reach if the one-sided 95% Clopper–Pearson bound $\text{UB}_{95} < 0.05/K$.

knockout base	K	pooled attempts	successes	UB_{95} vs $0.05/K$	certified
s0	23,322	1,491,420	0	$2.01 \times 10^{-6} < 2.14 \times 10^{-6}$	yes
s1	17,189	1,120,378	6	$1.06 \times 10^{-5} > 2.91 \times 10^{-6}$	no
s2	20,907	1,338,328	1	$3.55 \times 10^{-6} > 2.39 \times 10^{-6}$	no
s3	20,782	1,340,640	0	$2.24 \times 10^{-6} < 2.41 \times 10^{-6}$	yes
s4	21,554	1,360,222	3	$5.70 \times 10^{-6} > 2.32 \times 10^{-6}$	no
s5	22,297	1,419,656	2	$4.44 \times 10^{-6} > 2.24 \times 10^{-6}$	no

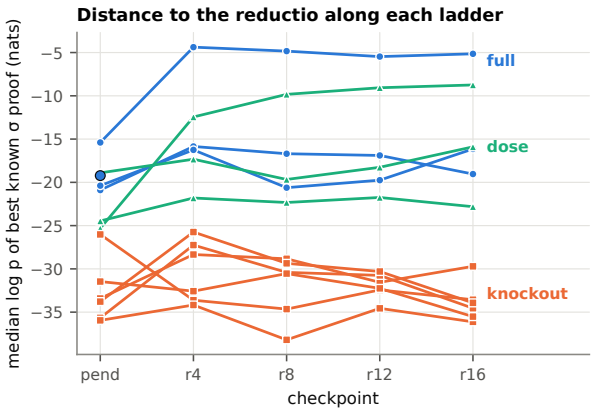


Figure 5: Median log-probability of the best known σ proof of the strict held-out members, along each ladder (one naming offset, $T = 0.8$). The full-exposure point at pend is seed 0's base.

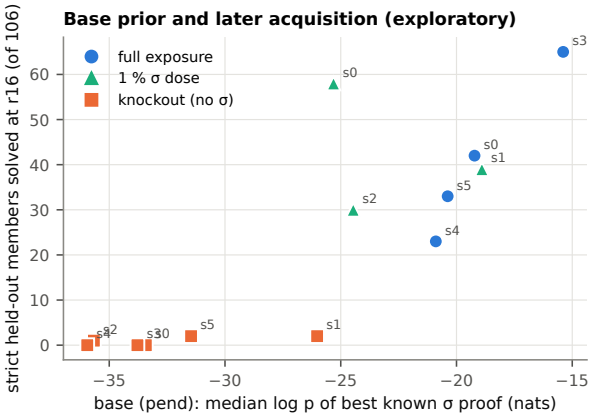


Figure 6: Exploratory: the base's median log-probability of known σ proofs against strict members solved at r16, for the 13 ladders with a scored base. Spearman $\rho = 0.88$ across all 13; 0.39 ($p = 0.38$) without the knockout arm.

member, $\hat{p}$ is about 10^{-6} to 3×10^{-4}. Knockout seed 1's base has 4 of its 18 strict converse-contraposition members within reach at its budget ($\hat{p} = 1.0$–$3.3 \times 10^{-4} \geq 1/17{,}189$). No family is certified out of reach for any seed: the largest K certifiable per family is below RL's budget.

In likelihood. We scored the best known σ proof of each of 101 strict held-out members under each model, teacher-forced at $T = 0.8$ (Fig. 5). Under the knockout bases the median log-probability is -26 to -36 nats; under full seed 0's base it is -19.2. The step that costs most is opening the box that assumes $\neg G$: -18.0 to -21.6 nats under the knockout bases, against -10.5 under the full base. Along the ladders, RL moves the full and dose models toward these proofs: from -15.4 to -5.1 nats for full s3, and from -25.3 to -8.7 for dose s0. It does not move the knockout models. Their medians at r16 are no closer. Four of six fall, and the largest rise is 3.7 nats (s0), against 10.3 nats for full s3 and 16.6 for dose s0. Two caveats apply. The scores use one naming offset, so they understate log-probability by at most $\ln 33 \approx 3.5$ nats. And the scored proofs are the full ladders' proofs, so a knockout model that found its own spelling (seed 1, below) need not raise them.

The base's prior predicts what RL acquires (exploratory). Across the 13 ladders with a scored base, the base's median log-probability of known σ proofs is rank-correlated with strict members solved at r16 (Spearman $\rho = 0.88$, Fig. 6). That is mostly the difference between arms

($\rho = 0.39$ without the knockout). Within a ladder, the base's log-probability of a member's best known proof predicts whether the member is solved at r16. The AUC is 0.92–0.95 on the four full seeds with scored bases, against a pre-registered bar of 0.7. It is 0.85–0.86 on the three dose seeds. Comparing only members of the same family, the AUC is 0.47–0.98 (full) and 0.28–0.60 (dose). It also predicts *when* a member is first solved in the ladder's per-round samples of held-out theorems. Spearman ρ between the base log-probability and the first-solve round is -0.69 to -0.87 on the full seeds and -0.61 to -0.72 on the dose seeds (unsolved members counted as round 17). Within families it averages -0.33 to -0.48 on the full seeds, against -0.22 to $+0.14$ on the dose seeds. The base's prior thus predicts which families RL acquires, and on the full seeds it also predicts the order within a family.

5.5 The one knockout ladder that moved

Knockout seed 1 was the only ladder whose free reductio kept spreading, so we continued it to round 24 (exploratory). Its free positive reductio proofs per round grew from 11 in round 16 to 123 in round 24. Its share of converse contraposition among strict training targets reached 0.33 from round 21. But its strict training targets stalled at 9, and the plain read solves the *same five* strict held-out members at r20 and r24: four converse-contraposition members and the generated theorem. It solves them more often. One member goes from 0 to 32 to 214 successes per 256 attempts at r16, r20 and r24. Three of its four converse-contraposition members (`la_transfer_1246`, `1460`, `2256`) are among the four its own base already had within reach at RL's budget. No family reaches the bar. We read this as RL eliciting a rare behaviour of this particular base, not creating a capability. Its r20 proofs of these members take three routes: σ (32 proofs), other σ^+ (24) and the lemma route (7).

5.6 What did not help, and what the knockout does not change

Seeding one proof. From each knockout r16 checkpoint we ran one more EI round, with and without four copies of knockout seed 1's base's own free-σ proof of the generated theorem. Seeded minus control was $+0/+0/+2/+0/+0$ strict family members on seeds s0, s1, s2, s3, s5 (s4 was not run). Seed 2's two were single successes in 256 attempts (one-sided exact $p = 0.25$). No family reached the bar. One extra round with a single seed proof is not enough.

Theorem-level reach. On the 322 evaluation theorems, RL's gain over what its own base reaches at RL's budget is similar in both arms. Each new base received 512 attempts, then at least $K_{\text{train}}(16)$ attempts on every theorem it had failed. The gain Δ_{cov} is $+37/+36/+32$ for full seeds s3–s5 and $+29/+31/+35/+37/+26/+21$ for the knockout seeds. RL's set-level advantage over its base does not depend on σ. These sets are not netted against RL-free controls.

A sharpening null for one sampler. Sharpening the base with no training is the strongest alternative to RL [15]. We implemented Metropolis–Hastings power sampling: target p^α at the sequence level, suffix resampling, 32 chains per theorem, blocks of 4 steps, 8 MH steps per block. Each theorem's plain-sampling comparator got exactly the same number of decoded tokens. On full seeds s0 and s1, the theorems RL r8 solves that the base cannot reach at K_{train} were 33 and 27. Power sampling at $\alpha = 4$ and $\alpha = 2$ solved 0/33 and 0/27. Plain sampling solved 1/33 and 1/27 at $T = 0.8$, and 1/33 and 2/27 at $T = 1.0$. On the knockout bases (s0, s1) power sampling solved 0/106 strict members, and plain sampling at $T = 1.0$ solved 1/106. The sampler fails its own positive control, however. On 17 theorems the base solves rarely, it solves 0/17 at $\alpha = 4$ and at

$\alpha = 1.5$, where plain sampling at the same tokens solves 6/17 ($T = 0.8$) and 13/17 ($T = 1.0$). The power runs also took 1.6–2.7 times the plain runs' wall-clock. The result is a null for this sampler at these settings, not for sharpening in general. That fits the finding that MCMC power samplers do not reach p^α at realistic budgets [35].

5.7 Robustness and checks

Lean. An independent reviewer re-checked with Lean 4.34.1 (the version used in the loop): every stored knockout proof of a strict member (448/448 accepted), samples from every arm, and all 16,961 refill records. The negative controls behaved as they should. None of 150 of the reads' own Lean rejections passed, none of 60 proofs under another theorem, none of 40 strict knockout proofs with `Classical.byContradiction` removed, and `sorry` failed. The pre-filter decided every rejection in knockout seed 1's 2.35 million distinct texts. Re-checking 1,950 pre-filter rejections (150 for each of 13 reasons) in Lean gave 0 accepted, a 95% upper bound of about 0.15% on the false-reject rate. Training never read an evaluation file. The training theorems share no renaming class with any evaluation set this paper reads, and none of the three pretraining sets shares one with the 106 strict members. Inherited from K12, the pretraining sets do share a few classes with general-ability sets, under a premise-order-invariant key: one textbook72 theorem, and 21–22 of the 5,000 held-out greedy theorems.

Cut-offs. An attempt is cut off when one step exceeds 512 tokens or the proof exceeds 96 steps. On the strict members at r16, 0.04–4.97% of knockout attempts were cut off, 0.11–2.89% of full attempts and 0.68–8.84% of dose attempts. A cut-off can only hide a success. The guided reads agree with the plain reads on every knockout and full seed. They cross the 0.1% cut-off line in 4 of 27 reads, against 64 of 74 plain reads. This paper adds a paired re-read with doubled caps (§H).

6 Test-time compute before and after RL

If RL mostly concentrates probability on proofs the base can already write, then spending more compute at test time on the base should recover part of RL's gain, and spending it on the RL model should add little. We measured two forms of test-time compute on the full-exposure seeds s0–s2. Neither comparison below is part of the knockout. Both are reviewed results of earlier runs (Fig. 7).

Guided step redraws. A guided read redraws a step that the logical checker rejects instead of abandoning the attempt (§3). At $k = 256$ it adds 18–22 holdout250 theorems to the base and -1 to $+4$ to the r8 and r16 models. The RL models are near the ceiling there, so textbook72 is the fairer pool. There guided reading adds $+8/+10/+12$ to the base, $+6/+4/+3$ at r8 and $+4/-1/+6$ at r16. On holdout250 it halves the gap between the base and r8, from 44/31/37 theorems (plain) to 21/14/19 (guided). The guided base solves 50–60% of the theorems RL "creates" at equal k. These reads are not compute-matched: a guided attempt costs about 1.8 times the tokens and 2.2 times the GPU time of a plain one. At matched sampled tokens, guided redraws still beat plain resampling on long textbook problems on all six r8 models tested (+5.2 to +10.3 points). The six are three seeds of this recipe and three of the same recipe trained on proofs of at most 6 lines. Guided reading does not reach most of the compute-matched creations, however. Of the theorems RL r8 solves that the base cannot reach at K_{pub} (22/21/14), the guided base solves only 3/4/1.

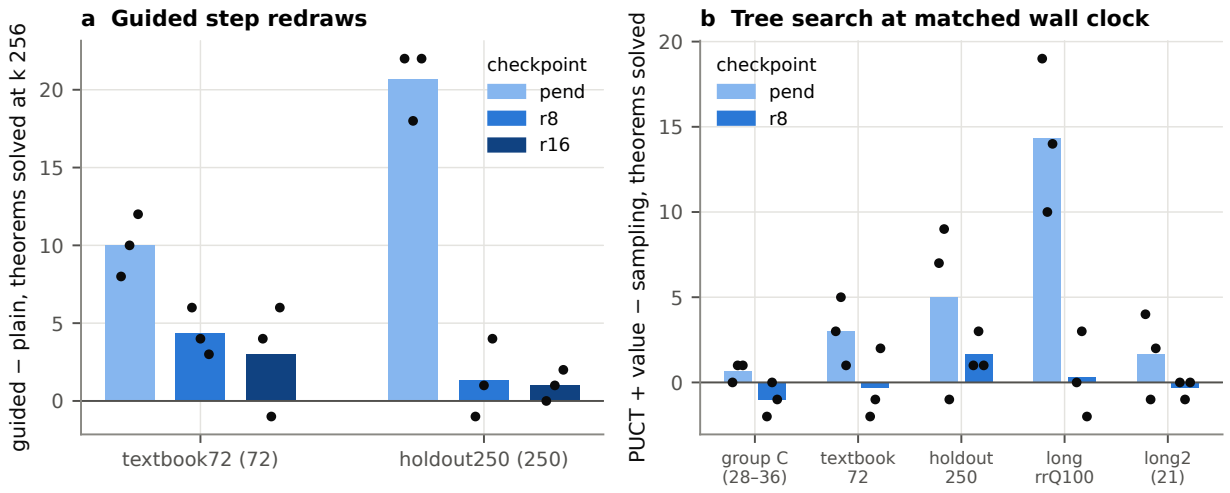


Figure 7: Test-time compute helps the base, not the RL model. (a) Guided minus plain reads, $k = 256$ each, on textbook72 and holdout250, for the base (pend) and after 8 and 16 rounds; bars are means, dots are seeds. Guided attempts cost about 2.2 times the GPU time of plain ones, so this is not compute-matched. On holdout250 the RL models sit within 3–14 theorems of the ceiling. (b) PUCT tree search with a learned value head minus sampling, at matched wall-clock on the same GPU; group C holds the theorems neither pend nor r8 solves in an earlier read (28–36 per seed).

Tree search. A PUCT search over proof states, with a value head trained on the frozen policy's rollouts, was compared with plain sampling at equal wall-clock on the same A40. At r8 it is within ±3 theorems of sampling on every pool. On group C, the theorems neither pend nor r8 solved in an earlier read, it solved 2/2/3 against sampling's 4/2/4. Its pre-registered gate ($\geq +3$ on two of three seeds) failed. A tenfold-longer exploratory read was also level (9/5/7 vs 10/7/5). At pend, search helps on a pool of 100 long theorems (13–16 lines): $+19/+10/+14$ over sampling. That is about 44% of what eight EI rounds add on this pool by sampling (+32.3). Of the +14.3 mean, +5.3 comes from the value head and the rest from search with the policy prior alone.

Reading. Both methods recover part of RL's gain at the base and add little after RL. That is what one expects if RL mostly converts reachable-but-unlikely proofs into likely ones. It also matches the likelihood view of §4, where RL raised the hardest step of its eventual proofs from about −6 to about −1 nats. Neither method reaches the theorems that a compute-matched base budget cannot reach. For those, step-level checking and search over the base's own proposals do not help, consistent with the knockout's message that what the base does not propose, RL rarely finds.

7 Discussion

What the knockout shows, and what it does not. The pre-registered verdict is that the classical families *need exposure*: from a σ-free base, this EI did not acquire them in 16 rounds, while every full-exposure seed did, and a 1% dose was enough. The verdict is about one algorithm and one horizon. It does not say that RL cannot produce σ. The knockout models wrote σ themselves, and one ladder spread it from round 9 onwards. A null result like this one has four readings, and the diagnostics bear on each.

1. *σ is out of the policy's reach.* This is false for four of six bases, which write σ proofs of family members at large k. It cannot be excluded for the two certified bases (s0, s3) at RL's budget, though even there it is not excluded at larger budgets.
2. *EI explores too little.* EI spends 32 attempts per training theorem per round. On the 123 strict training targets that is 3,936 attempts per round. At the knockout bases' pooled rates (0 to 5.4×10^{-6} per attempt on held-out members) it yields at most 0.02 expected σ proofs per round, or at most 0.34 over 16 rounds. Most knockout ladders therefore had little or nothing to train on. The three ladders that solved any strict training target by round 16 (s1, s2, s5: 5, 1 and 2 targets) are the three whose ladders wrote the free reductio earliest (rounds 9, 12 and 2).
3. *EI amplifies rare finds too slowly.* Knockout seed 1 multiplied its free reductio more than tenfold in eight extra rounds, yet its held-out set stayed at 5 members. Amplification there was real but narrow: the model became reliable on the members it could already reach, without generalising to the family. A single seeded proof did nothing in one round. The test of seeding at volume is EI with the 1,550 dose records in replay instead of pretraining, and it is not run.
4. *The knockout made the bases worse learners in general.* This is not supported. RL's compute-matched gain on the 322 evaluation theorems is the same in both arms (+21 to +37), and the knockout's general-ability cost sits in theorems that need σ.

Readings 2 and 3 together describe an *amplification threshold*. EI turns a composition the base produces at moderate rates into a capability within a few rounds, and it does nothing visible with one the base produces at 10^{-6}. The dose arm locates the threshold between 0% and 1% of pretraining records for this organism and budget, and the 0.1% arm would narrow it.

Relation to controlled studies. Tsilivis et al. [37] show, on parity, that RL after next-token pretraining amplifies a rare long-reasoning demonstration efficiently as long as its share of the data is not exponentially small. Our 1% dose sits on the efficient side. Our knockout leaves the composition at about 10^{-6} per attempt, which for EI at $k = 32$ behaves like "exponentially small". Zero exposure was not tested there, as far as the quotes we verified go. Zhang et al. [50] find, in a 100M model trained from scratch, that 0% or 0.1% exposure to a context in pretraining gives no transfer after RL and 1% enhances it. That is close to our dose pattern, for a context rather than an inference rule. Abdulsalam et al. [1] remove shortcuts from pretraining in a rewrite grammar and report RL solving buckets on which the pretrained policy scores 0% at pass@1024. In our organism, a pass@1024 of zero is compatible with per-attempt rates of 10^{-6}, which a million attempts reveal. Their new behaviour is a new action type, while ours is a composition of two rules the base already uses. More importantly, the algorithm that created it there was GRPO. Rejection-sampling fine-tuning, the family our EI belongs to, plateaued, and they attribute the difference to selectivity rather than exploration volume. Our null may therefore be specific to EI. The two results are not in conflict, but neither settles the other. A GRPO arm from the knockout bases is the obvious test.

An evidence ladder for "RL created it". The measurement half of the paper suggests a reporting standard, in increasing order of strength:

1. equal-k success;
2. a compute-matched base budget, reported as a curve over defensible allocations;
3. RL-free controls with matched training;
4. a Clopper–Pearson certificate against the base at that budget;
5. an intervention on what the base saw.

In our full-exposure seeds the claim shrinks at each step: 54/51/60 theorems at equal k, 21/20/14 under every allocation, 20/14/8 net of replay at K_{pub}, and 4 (theorem, seed) pairs on 3 theorems

surviving every control, of which one theorem is classical-only. The intervention then shows that, for one composition, exposure in pretraining decides whether this RL acquires it. Each step is cheap in a small organism and expensive in a large one. Steps 2 and 3 are affordable at any scale and are rarely reported.

Implications for evaluation and data filtering. Two practical points follow, with the caveat that they come from a small organism. First, a capability that is absent at $k = 256$ or $k = 1{,}024$ can still be present at 10^{-6} per attempt, and RL is the kind of process that can amplify such rates. Evaluations that aim to bound a capability need rare-event estimates, such as large-k sampling, likelihoods of known solutions or importance sampling, not pass@k at modest k. Second, the knockout is a data-filtering intervention. Removing a composition from pretraining prevented its acquisition by this RL within this budget, but 1% leakage undid the filter. A filter that leaves even a small fraction of the composition in place may not slow later RL at all.

Interface. An earlier, smaller organism of the same project, a 3.2M-parameter model trained on proofs of at most six lines, had 29 theorems that expert iteration solved and that one base seed solved 0 times in 400,000 whole-proof attempts. A base of the same size acting through the proof-state interface reaches 28 of them (at least one accepted proof in up to 400,000 attempts; at EI's own bar of $\hat{p} \geq 0.01$ it clears 21 and 14 on its two seeds). There were only two seeds per arm, and the state cannot be separated from step-by-step generation. Whether RL "created" something can depend on the interface through which the base is asked. We leave that axis to future work, at the 9.56M scale.

8 Limitations

- **Scale and domain.** One architecture with 9.56M parameters, propositional logic over at most four atoms, and pretraining proofs of at most 12 lines. Nothing here is measured on a large language model or on first-order logic. The organism was chosen because it allows interventions that large models do not. The price is that results about it are results about it.
- **One RL algorithm, 16 rounds.** T1 expert iteration with $k = 32$ at $T = 0.8$ is rejection-sampling fine-tuning. GRPO-style updates, more exploration (higher temperature, larger k on failed targets) or longer ladders could behave differently. In the closest controlled study, GRPO built behaviours absent from pretraining while rejection-sampling fine-tuning plateaued [1]. An earlier comparison from our own full-exposure checkpoints found GRPO with a pass@4 advantage ahead of EI on the hardest theorems, though mostly through one seed. "Needs exposure" is a statement about 16 rounds of this EI, and one knockout seed was still changing at round 24.
- **Seeds and power.** Six seeds per arm power the primary indicator but not most per-family tests. Three families are 3 vs 0 (Holm 0.17), and excluded middle, acquired by one full seed, is untestable at these counts. A 0/6 outcome bounds the per-seed rate of creation below 0.39 only; creation in one seed out of ten would likely be missed. The dose arm has three seeds.
- **Near-saturated training targets.** By round 8 the full ladders solve 97–98% of the 4,495 training theorems. The classical families are a small part of the training signal, and the classical held-out members are hard. A pool richer in classical targets might change the dynamics.
- **The knockout is not surgical.** Besides σ, it lowers the share of `DN` and ¬-introduction records and the share of classical-only theorems. The knockout bases are 6.88 points worse at greedy

held-out decoding (all of it on theorems that need σ) and solve fewer intuitionistic textbook72 theorems at pend (-4.0 of 53, three full seeds). The textbook72 gap persists at r16.

- **"Absent" is not established.** Only 2 of 6 knockout bases are certified out of reach at RL's budget, and no family is certified for any seed. The estimator that would quantify smaller rates runs two- to four-fold low below 10^{-4} and was calibrated on full-exposure models. The likelihood scores use the full ladders' proofs and one naming offset.
- **Sharpening.** The power-sampling null holds for one Metropolis–Hastings sampler that fails its own positive control. It says nothing about p^α itself.
- **Cut-offs and settings.** Many plain reads cut off more than 0.1% of attempts in some stratum (up to 8.8% of strict-member attempts on one dose seed). We re-read the strict members with doubled caps (§H). The old and new full seeds differ in GPU class (RTX A6000 vs A40) and sampling batch for rounds 1–8, and are pooled. The budget conversions rest on one measured class ratio.
- **Measurement results are on three seeds.** §4 uses the three earlier full-exposure seeds. Its test-time-compute comparisons with guided reads are not compute-matched.

9 Reproducibility

Code, data and models. All code is in the fork `dan-pandori/nd-takehome`. The experiments are on branch `dan_create-or-elicit` (commit `8e195390`, with the review): scripts `coe/` and `pod/coe/`, the reviewer's independent recount code `review_coe/`, the pre-registration, the per-round statistics, the budget and estimator outputs, and the claims ledger. This paper is on `dan_create-or-elicit-paper`. Its figures and statistics are regenerated by `paper/scripts/` from those files, and every number in the text has a row in `paper/LEDGER.md` pointing to its reviewed source. The pretraining sets, all 200 ladder checkpoints, the read rows, the large-k samples and the Lean-gate dumps are in the public bucket `hf://buckets/dan-pandori/nd-rl/create-or-elicit/`. The decompressed md5 sums of the pretraining sets are K12 `800b5486`, K12$^-$ `466fc214` and K12$^{1\%}$ `19cd2263`.

Pre-registration and review. The experiment's pre-registration was committed before the first GPU pod was created. Six dated addenda each precede the work they govern, with one disclosed exception. Addendum 2 replaced the held-out sets with the strict sets after the first round of the first knockout ladder. The reviewer's recount shows the original sets give the same primary result. An independent reviewer re-derived every count in §5 blind to the write-up, from the raw files and with separate code, and re-checked the proofs in Lean. The reviewer corrected three Boschloo p-values computed on a transposed table (no verdict changed) and asked for ten rewordings. This paper uses the corrected values and the reviewed wording. This writing run was pre-registered as well (`preregistration/create-or-elicit-paper.md`), including the checks of Appendix H.

Compute. Table 6 gives the compute per arm. Every 16-round ladder made 2,301,440 attempts on training theorems and 9,600 fine-tuning steps. The knockout and dose ladders' compute is within 0.95–1.03 times the full ladders' on every quantity. The experiments used 16 pods, 15 NVIDIA A40 ($0.59/h) and one RTX A6000: 309.34 pod-hours and $182.50 in total. The earlier runs whose results §4 and §6 quote record their compute in their own reports (Appendix I).

Table 6: Compute per new 16-round ladder, mean over seeds (A40; results-registry rows, recounted by the reviewer). Pretraining took a further ≈1,280 GPU-seconds, ≈24,000 steps and $\approx 4.4 \times 10^8$ tokens per seed.

quantity	KNOCKOUT	FULL (s3–s5)	DOSE
sampling GPU-s	32,441	34,005	34,492
evaluation GPU-s	7,886	8,046	7,970
fine-tuning GPU-s	13,747	13,753	13,898
total GPU-s	54,105	55,841	56,390
generated tokens	6.69×10^8	6.73×10^8	6.76×10^8
Lean checks	2.15×10^6	2.21×10^6	2.18×10^6

Acknowledgements

Left for Dan.

References

[1] Azwar Abdulsalam, Nishil Patel, and Andrew Saxe. RL Post-Training Builds Compositional Reasoning Strategies. arXiv:2607.07646, 2026. Version read: v1. arXiv comments: accepted to the 2nd Workshop on Compositional Learning at ICML 2026.

[2] Thomas Anthony, Zheng Tian, and David Barber. Thinking Fast and Slow with Deep Learning and Tree Search. arXiv:1705.08439, 2017. Version read: v4.

[3] Kshitij Bansal, Christian Szegedy, Markus N. Rabe, Sarah M. Loos, and Viktor Toman. Learning to Reason in Large Theories without Imitation. arXiv:1905.10501, 2019. Version read: v3.

[4] Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, and Azalia Mirhoseini. Large Language Monkeys: Scaling Inference Compute with Repeated Sampling. arXiv:2407.21787, 2024. Version read: v3.

[5] Zachary Burton. Inference-Time Diversity in RL-Trained Lean Theorem Provers: A Diagnostic Study. arXiv:2601.16172, 2026. Version read: v3.

[6] Tom Davidson, Jean-Stanislas Denain, Pablo Villalobos, and Guillem Bas. AI capabilities can be significantly improved without expensive retraining. arXiv:2312.07413, 2023. Version read: v1.

[7] Aghyad Deeb and Fabien Roger. Do Unlearning Methods Remove Information from Language Model Weights? arXiv:2410.08827, 2024. Version read: v3.

[8] Kefan Dong and Tengyu Ma. STP: Self-play LLM Theorem Provers with Iterative Conjecturing and Proving. arXiv:2502.00212, 2025. Version read: v4.

[9] Ryan Greenblatt, Fabien Roger, Dmitrii Krasheninnikov, and David Krueger. Stress-Testing Capability Elicitation With Password-Locked Models. arXiv:2405.19550, 2024. Version read: v1.

[10] Felix Hofstätter, Teun van der Weij, Jayden Teoh, Rada Djoneva, Henning Bartsch, and Francis Rhys Ward. The Elicitation Game: Evaluating Capability Elicitation Techniques. In *International Conference on Machine Learning (ICML)*, 2025. Version read: arXiv v3.

[11] Shengding Hu, Xin Liu, Xu Han, Xinrong Zhang, Chaoqun He, Weilin Zhao, Yankai Lin, Ning Ding, Zebin Ou, Guoyang Zeng, Zhiyuan Liu, and Maosong Sun. Predicting Emergent Abilities with Infinite Resolution Evaluation. arXiv:2310.03262, 2023. Version read: v3.

[12] Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P. Dick, Hidenori Tanaka, Edward Grefenstette, Tim Rocktäschel, and David Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. arXiv:2311.12786, 2023. Version read: v2.

[13] Xiaotong Ji, Rasul Tutunov, Matthieu Zimmer, and Haitham Bou Ammar. Scalable Power Sampling: Unlocking Efficient, Training-Free Reasoning for LLMs via Distribution Sharpening. arXiv:2601.21590, 2026. Version read: v1 (abstract only).

[14] Erik Jones, Meg Tong, Jesse Mu, Mohammed Mahfoud, Jan Leike, Roger Grosse, Jared Kaplan, William Fithian, Ethan Perez, and Mrinank Sharma. Forecasting Rare Language Model Behaviors. arXiv:2502.16797, 2025. Version read: v1.

[15] Aayush Karan and Yilun Du. Reasoning with Sampling: Your Base Model is Smarter Than You Think. arXiv:2510.14901, 2025. Version read: v1.

[16] Joshua Kazdan, Rylan Schaeffer, Youssef Allouah, Colin Sullivan, Kyssen Yu, Noam Levi, and Sanmi Koyejo. Efficient Prediction of Pass@k Scaling in Large Language Models. arXiv:2510.05197, 2025. Version read: v1.

[17] Tomasz Korbak, Ethan Perez, and Christopher L. Buckley. RL with KL penalties is better viewed as Bayesian inference. In *Findings of EMNLP 2022*, 2022. Version read: arXiv v2.

[18] Akshay Krishnamurthy, Audrey Huang, and Nived Rajaraman. Select and Improve: Understanding the Mechanics of Post-Training for Reasoning. arXiv:2606.13125, 2026. Version read: v1.

[19] Guillaume Lample, Marie-Anne Lachaux, Thibaut Lavril, Xavier Martinet, Amaury Hayat, Gabriel Ebner, Aurélien Rodriguez, and Timothée Lacroix. HyperTree Proof Search for Neural Theorem Proving. arXiv:2205.11491, 2022. Version read: v1.

[20] Yuhao Li and Shengchao Liu. On Distinguishing Capability Elicitation from Capability Creation in Post-Training: A Free-Energy Perspective. arXiv:2605.08368, 2026. Version read: v1 (abstract, Sec. 1, 2.3, 4).

[21] Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. ProRL: Prolonged Reinforcement Learning Expands Reasoning Boundaries in Large Language Models. arXiv:2505.24864, 2025. Version read: v1.

[22] Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: evaluating LLMs with fewer examples. In *International Conference on Machine Learning (ICML)*, 2024. Version read: arXiv v2.

[23] Simon Park, Simran Kaur, and Sanjeev Arora. How Does RL Post-training Induce Skill Composition? A Case Study on Countdown. arXiv:2512.01775, 2025. Version read: v1.

[24] Stanislas Polu and Ilya Sutskever. Generative Language Modeling for Automated Theorem Proving. arXiv:2009.03393, 2020. Version read: v1 (method and selected tables).

[25] Stanislas Polu, Jesse Michael Han, Kunhao Zheng, Mantas Baksys, Igor Babuschkin, and Ilya Sutskever. Formal Mathematics Statement Curriculum Learning. arXiv:2202.01344, 2022. Version read: v1 (method, results, discussion).

[26] Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-Tuning Enhances Existing Mechanisms: A Case Study on Entity Tracking. arXiv:2402.14811, 2024. Version read: v1.

[27] Yuval Ran-Milo, Yotam Alexander, Shahar Mendel, and Nadav Cohen. Outcome-Based RL Provably Leads Transformers to Reason, but Only With the Right Data. arXiv:2601.15158, 2026. Version read: v4 (abstract only).

[28] Rylan Schaeffer, Joshua Kazdan, John Hughes, Jordan Juravsky, Sara Price, Aengus Lynch, Erik Jones, Robert Kirk, Azalia Mirhoseini, and Sanmi Koyejo. How Do Large Language Monkeys Get Their Power (Laws)? arXiv:2502.17578, 2025. Version read: v1.

[29] Rylan Schaeffer, Noam Levi, Brando Miranda, and Sanmi Koyejo. Pretraining Scaling Laws for Generative Evaluations of Language Models. arXiv:2509.24012, 2025. Version read: v2.

[30] Lisa Schut, Nenad Tomasev, Tom McGrath, Demis Hassabis, Ulrich Paquet, and Been Kim. Bridging the Human-AI Knowledge Gap: Concept Discovery and Transfer in AlphaZero. arXiv:2310.16410, 2023. Version read: v1.

[31] Rulin Shao, Shuyue Stella Li, Rui Xin, Scott Geng, Yiping Wang, Sewoong Oh, Simon Shaolei Du, Nathan Lambert, Sewon Min, Ranjay Krishna, Yulia Tsvetkov, Hannaneh Hajishirzi, Pang Wei Koh, and Luke Zettlemoyer. Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947, 2025. Version read: v2 (abstract only).

[32] Jingyan Shen, Ang Li, Salman Rahman, Yifan Sun, Micah Goldblum, Matus Telgarsky, and Pavel Izmailov. Understanding Reasoning from Pretraining to Post-Training. arXiv:2607.16097, 2026. Version read: v2.

[33] David Silver, Thomas Hubert, Julian Schrittwieser, Ioannis Antonoglou, Matthew Lai, Arthur Guez, Marc Lanctot, Laurent Sifre, Dharshan Kumaran, Thore Graepel, Timothy Lillicrap, Karen Simonyan, and Demis Hassabis. Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm. arXiv:1712.01815, 2017. Version read: v1 (targeted sections).

[34] Wenhe Sun, Cunxiang Wang, Zijun Yao, and Yixin Cao. From Base Rollouts to RL Reasoning: A Budgeted Search Perspective. arXiv:2609.01274, 2026. Version read: v1.

[35] Abiel Malkani Talwar, Christopher Nemeth, and Henry Moss. A Critical Look at Power Sampling when Test Time Sharpening LLMs. arXiv:2610.13653, 2026. Version read: v1.

[36] Sang Truong, Yuheng Tu, Rylan Schaeffer, and Sanmi Koyejo. Item Response Scaling Laws: A Measurement Theory Approach for Efficient and Generalizable Neural Scaling Estimation. arXiv:2606.07616, 2026. Version read: v1.

[37] Nikolaos Tsilivis, Eran Malach, Karen Ullrich, and Julia Kempe. How Reinforcement Learning After Next-Token Prediction Facilitates Learning. arXiv:2510.11495, 2025. Version read: v2.

[38] Teun van der Weij, Felix Hofstätter, Oliver Jaffe, Samuel F. Brown, and Francis Rhys Ward. AI Sandbagging: Language Models can Strategically Underperform on Evaluations. arXiv:2406.07358, 2024. Version read: v4.

[39] Constantin Venhoff, Iván Arcuschin, Philip Torr, Arthur Conmy, and Neel Nanda. Base Models Know How to Reason, Thinking Models Learn When. arXiv:2510.07364, 2025. Version read: v4.

[40] Siwei Wang, Yifei Shen, Haoran Sun, Shi Feng, Shang-Hua Teng, Li Dong, Yaru Hao, and Wei Chen. Benefits and Pitfalls of Reinforcement Learning for Language Model Planning: A Theoretical Perspective. arXiv:2509.22613, 2025. Version read: v2.

[41] Fang Wu, Weihao Xuan, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The Invisible Leash: Why RLVR May or May Not Escape Its Origin. arXiv:2507.14843, 2025. Version read: v4.

[42] Gabriel Wu and Jacob Hilton. Estimating the Probabilities of Rare Outputs in Language Models. arXiv:2410.13211, 2024. Version read: v2.

[43] Mingqi Wu, Zhihao Zhang, Qiaole Dong, Zhiheng Xi, Jun Zhao, Senjie Jin, Xiaoran Fan, Yuhao Zhou, Huijie Lv, Ming Zhang, Yanwei Fu, Qin Liu, Songyang Zhang, and Qi Zhang. Reasoning or Memorization? Unreliable Results of Reinforcement Learning Due to Data Contamination. arXiv:2507.10532, 2025. Version read: v3 (abstract only). arXiv comments: AAAI 2026.

[44] Zijian Wu, Suozhi Huang, Zhejian Zhou, Huaiyuan Ying, Zheng Yuan, Wenwei Zhang, Dahua Lin, and Kai Chen. InternLM2.5-StepProver: Advancing Automated Theorem Proving via Critic-Guided Search. arXiv:2410.15700, 2024. Version read: v2.

[45] Huajian Xin, Z. Z. Ren, Junxiao Song, Zhihong Shao, Wanjia Zhao, Haocheng Wang, Bo Liu, Liyue Zhang, Xuan Lu, Qiushi Du, Wenjun Gao, Qihao Zhu, Dejian Yang, Zhibin Gou, Z. F. Wu, Fuli Luo, and Chong Ruan. DeepSeek-Prover-V1.5: Harnessing Proof Assistant Feedback for Reinforcement Learning and Monte-Carlo Tree Search. arXiv:2408.08152, 2024. Version read: v1.

[46] Ran Xin, Chenguang Xi, Jie Yang, Feng Chen, Hang Wu, Xia Xiao, Yifan Sun, Shen Zheng, and Kai Shen. BFS-Prover: Scalable Best-First Tree Search for LLM-based Automatic Theorem Proving. arXiv:2502.03438, 2025. Version read: v3.

[47] Chen Yang and Jun Chen. RLVR is a Kernel, Not a Function: Statistical Inference for pass@k Crossovers. arXiv:2609.22547, 2026. Version read: v2.

[48] Haohui Yang, Jiaxing Sun, and Xiujun Ma. More Correct Mass, Worse Answers: Why Power Sampling Can Fail and How to Fix It. arXiv:2608.14420, 2026. Version read: v1.

[49] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model? arXiv:2504.13837, 2025. Version read: v5.

[50] Charlie Zhang, Graham Neubig, and Xiang Yue. On the Interplay of Pre-Training, Mid-Training, and RL on Reasoning Language Models. arXiv:2512.07783, 2025. Version read: v1.

A Glossary

pend, rN
the model at the end of pretraining; after N rounds of expert iteration.

plain read / guided read
$k = 256$ independent attempts per theorem at $T = 0.8$ / the same with logically rejected steps redrawn (at most 10 per attempt). x0, x1: two sample seeds of a plain read.

σ, σ^+, lemma route
`DN` on a line built by $\neg$-introduction / `DN` on any derived line / proving $\neg\neg G \to G$ inside an implication box and applying it.

free positive reductio
`DN` on a derived line whose conclusion G is not a negation and whose $\neg\neg G$ is not a subformula of the theorem, by any route.

strict member
a family member that is classical-only (no intuitionistic proof, by G4ip) and has no $\neg\neg$ subformula in its sequent.

at the bar
a family with at least 50% of its strict held-out members solved in a plain read.

within reach at K
the base's empirical success rate $\hat{p} \geq 1/K$ over at least K attempts.

certified out of reach at K
one-sided 95% Clopper–Pearson upper bound $< 0.05/K$.

Δ_{cov}
theorems RL solves at $k = 256$ minus theorems the base reaches within K.

B Construction of the knockout set

K12$^-$ removes every K12 record whose written or pruned proof applies `DN` to a derived double negation (16,961 records: 16,703 σ and 258 whose `DN` cites an $\to$-elimination, $\vee$-elimination, $\bot$-elimination, `DN` or $\wedge$-elimination line). It refills 16,961 records, drawn uniformly per length from K12's own unused source pools: lengths up to 6 from the depth-3 generator pool and lengths 7–12 from the long-proof pool. Every refill record passes every K12 filter (no box depth ≥ 3; `DN` only on assumptions or premises) and is σ^+-free and lemma-route-free. Under an order-sensitive and a premise-order-invariant renaming key, the refill shares no class with K12 or with any of the 100 data and target files of the repository. One of K12's 11 exclusion pools was absent when the set was built and could not be checked. Every refill record's proof, translated to Lean, is accepted (16,961/16,961). Unchanged proofs pass the same harness, and goal-negated proofs fail it. Removal and refill leave the per-length counts identical. The dose set K12$^{1\%}$ swaps 1,550 refill records for removed σ records at K12's σ length mix. A 0.1% set is built but unused.

C When each ladder first trained on each family

First round in which at least half of a family's strict training targets had an accepted proof in the ladder (every target receives 32 attempts per round; "–" = not within 16 rounds).

ladder	contrap. conv.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan
FULL s0/s1/s2	1/2/1	4/13/3	–/–/3	–/–/5	–/14/–	never
FULL s3/s4/s5	2/1/1	2/–/–	8/–/–	3/–/16	never	never
KNOCKOUT s0–s5	never	never	never	never	never	never
DOSE s0/s1/s2	2/1/3	3/7/7	12/–/–	7/–/–	never	never

D Budgets at r16 and the estimator

At r16 the five allocations are $K_{\text{att}} = 7{,}147$; $K_{\text{train}} = 17{,}531/15{,}884/20{,}422$; $K_{\text{uni}} = 24{,}619/22{,}827/29{,}510$; $K_{\text{pub}} = 47{,}962/49{,}428/50{,}447$; $K_{\text{adapt}} = 109{,}875/100{,}929/119{,}575$ base attempts per theorem. Because K_{adapt} exceeds the 66,304 attempts most hard theorems received, the r16 sets at K_{adapt} are mostly undetermined. Figure 8 shows the known-proof estimator's calibration.

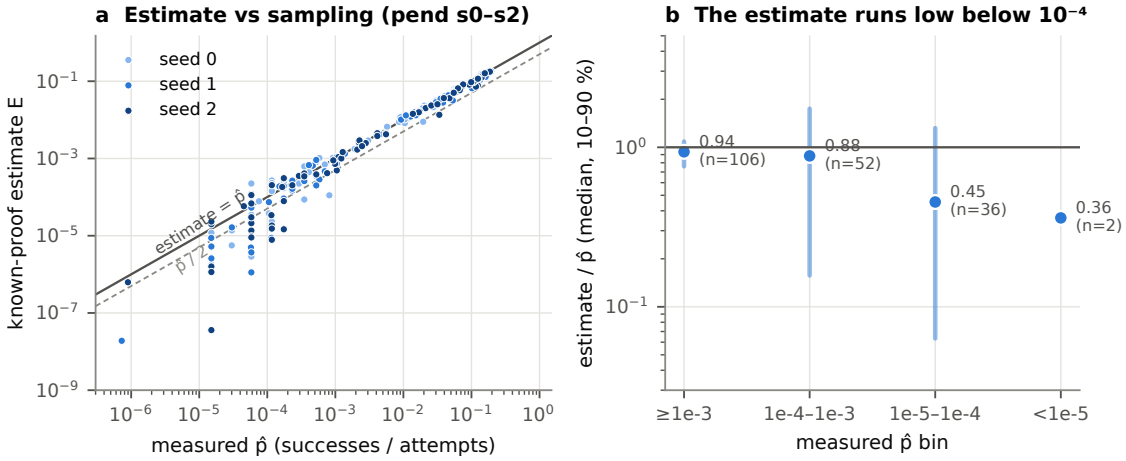


Figure 8: Known-proof estimator, full-exposure bases s0–s2. (a) Estimate against measured $\hat{p}$ for (seed, theorem) pairs with at least one success in large-k sampling. (b) Median and 10–90% range of $E/\hat{p}$ by $\hat{p}$ bin.

E Power sampling

The sampler targets p^{α} over whole proofs by Metropolis–Hastings with suffix resampling (`coe/power_sample.py`): 32 chains per theorem, blocks of 4 proof steps, 8 MH steps per block (32 in the convergence check), $\alpha \in \{4, 2\}$ (and 1.5 in the control). Acceptance rates were 0.55–0.74. On a development pilot the per-chain success was 0.752 at 8 MH steps per block and 0.760 at 32. The plain comparators received exactly the decoded tokens of the power runs. They store no proof text, so their 1–2 solves could not be re-checked in Lean.

F Pre-registered expectations against outcomes

The experiment's pre-registration stated numeric expectations. As tallied by the reviewer, they *hit* on:

- the shape of K12^{-};
- the general-ability gap on held-out greedy decoding and on non-σ theorems;

- 0/123 strict targets in round 1 and 0/106 held-out members at pend on all knockout seeds;
- knockout median log-probabilities at or below -15 nats;
- converse contraposition acquired on 3/3 new full seeds;
- at least one family on 6/6 full and 0/6 knockout seeds;
- the primary p and the arm medians;
- recovery of non-classical holdout250 by r16;
- $\Delta_{\text{cov}} > 0$;
- the power-sampling nulls;
- the dose arm's round-1 σ proofs and family acquisition;
- knockout seed 1's extension.

They *missed* on:

- the classical-only share falling by $\geq 70\%$ (observed 67.6%);
- textbook72 within ± 4 (observed -6.5);
- certificates on $\geq 5/6$ knockout bases (observed 2/6);
- negated conditional on $\geq 2/3$ new full seeds (observed 1/3);
- the first σ^+ proof after round 4 (it appeared in round 1, on theorems supplying $\neg\neg G$);
- Δ_{cov} within $+8$ to $+35$ (two seeds at $+36$ and $+37$);
- 0/106 for every sampler on the knockout bases (plain sampling at $T = 1$ solved one);
- 3–15 strict targets in the dose arm's round 1 (observed 2/17/2);
- a seeded-round gain on ≥ 3 seeds (observed 1/5).

G Corrections applied

The experiment's own scripts passed two-by-two tables to `scipy.stats.boschloo_exact` with the groups as rows. That function treats columns as groups. With the arms as columns, the RL-only test is $p = 0.0032$ (reported 0.0020), the excluded-middle test 0.39 (0.30) and the three-versus-none family tests 0.034 (0.035). The primary $p = 0.00024$ and the dose test $p = 0.0033$ are unchanged. `paper/scripts/stats_extra.py` recomputes all five. The reviewer's ten rewordings are applied throughout. The most important ones: "need" becomes "not acquired by this EI within 16 rounds", "reach" is used only in its technical sense, and the sharpening null is stated for one sampler.

H This paper's checks

[Pending: the pre-registered checks C1–C3 are running; results will be filled in here.]

I Provenance of earlier results

Run names refer to reviewed reports in the fork `dan-pandori/nd-takehome` (branch `dan_<run>`, file `review_<run>.md`) and to their summaries in `nd-rl/experiment-summaries`.

- create-or-elicit: the knockout, dose and full s3–s5 ladders; Phase 0 (budgets, estimator) and Phase 2 (power sampling). Review commits `dd8ee82e` (blind recount), `8e195390` (verdict).
- capability-defs: the measurement framework on full seeds s0–s2 (budgets, certificates, controls, definitions, guided reads). Review `da11be78`.
- trajectory, rl-continue: full seeds s0–s2, rounds 1–8 and 9–16; exact likelihoods along RL.
- rl-from-ckpt: the replay-only control.
- guided-tts: guided redraws at matched tokens; the step checker.
- mcts-a: PUCT search with a learned value (review `5e70afbd`).

- grpo-best: GRPO variants from the same checkpoints.
- support-curves, support-state, claim-audit: the 3.2M organism and the interface confound.

The claims ledger (`claims_ledger.md`, rows L1–L101) maps every inherited number to its reviewed source.